

Assignment 2

Consider a non-relativistic particle moving under the influence of an electromagnetic field. Its equations of motion are

$$\frac{d\mathbf{r}}{dt} = \mathbf{v} \quad (1)$$

$$\frac{d\mathbf{v}}{dt} = \frac{q}{m}(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2)$$

Write a program that can solve for the particle motion (i.e. particle position and velocity as a function of time) given any general form of electromagnetic field (i.e., define functions for electric and magnetic fields so that they can be easily changed). This program should use the RK4 method. It should take as an input the starting position and velocity of the particle. Write a code for 3D geometry.

Suppose that a particle has $q = 1$ and $m = 1$, and it is launched from $(x, y, z) = (0, 0, 0)$ with an initial velocity $\mathbf{v} = 0.1\hat{\mathbf{y}}$. Solve it for the specific case where the electric field is $\mathbf{E} = 0.02\hat{\mathbf{y}}$ and magnetic field is $\mathbf{B} = (1 + 0.1x + 0.01t)\hat{\mathbf{z}}$. The program should output a figure showing the trajectory of the particle in the $x - y$ plane from $t = 0$ to $t = 100$. The plot should be properly labelled. Submit a file with the source code and the plot.

Tip: First test your code for simple cases like a uniform magnetic field or a uniform electric field.